THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 12: **How Nature Works in Harmony** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Removing top carnivores from a forest can harm the whole ecosystem. Reason (R): Herbivore numbers increase and overgraze the vegetation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A food chain has producers with 50000 J. If it has 5 levels (producers, primary, secondary, tertiary, quaternary), the energy at the quaternary level is: 1

(a) 50 J **(b)** 500 J **(c)** 5 J **(d)** 0.5 J

3. Which of the following is a good example of mutualism? 1

(a) tapeworm and human **(b)** tiger and deer
(c) mosquito and human **(d)** bees and flowers

4. Assertion (A): Energy flow in an ecosystem is unidirectional. Reason (R): Energy lost as heat cannot be reused by producers. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. In a sample ecosystem, 1 kg of tertiary consumer needs about how many kg of producers, using 10 per cent rule on biomass? 1

(a) 10000 kg **(b)** 100 kg **(c)** 10 kg **(d)** 1000 kg

6. The relationship between a tapeworm and its host is: 1

(a) predation **(b)** competition **(c)** mutualism **(d)** parasitism

7. Assertion (A): Food chains rarely have more than four or five trophic levels. Reason (R): About 90 per cent of energy is lost at each trophic level. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. If a pesticide kills all the frogs in a field, the number of insects will most likely: 1

(a) stay zero **(b)** increase **(c)** decrease **(d)** become constant

9. Lichen is an association of: 1

(a) bacteria and plants **(b)** algae and fungi

(c) fungi and animals **(d)** algae and animals

10. Assertion (A): Decomposers are essential for the recycling of nutrients. Reason (R): They convert dead organic matter into simple inorganic substances. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Calculate the energy available to the primary consumer, secondary and tertiary if producers have 100000 J (10 per cent rule). 2

12. How do forests act as carbon sinks? 2

13. Why do herbivores form the largest biomass in animals but fewer in number than plants? 2

14. Why can heavy use of pesticides in fields harm the birds that feed on insects there? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A pond has 1000 kg of algae. By the 10 per cent biomass rule, find the maximum mass of tertiary consumers it can support. 3

16. Describe the effects of eutrophication in a lake in three stages. 3

17. Why do some scientists recommend reducing meat consumption? Use energy ideas. 3

OR

A farmer sprays an insecticide that kills all insects in a field. Predict three consequences. 3

18. State how the afforestation of degraded land restores the ecosystem. 3

OR

What happens to an ecosystem when the number of decomposers increases greatly in a pond rich in organic waste? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The population of lions in a forest dies due to disease. Trace the consequences through the food web with at least five steps in the ecosystem and the final outcome. 5

OR

Discuss how rising CO2 in the atmosphere affects plants, oceans, ice and weather, and what individuals can do. 5

20. Analyse the food web: grass, grasshopper, rabbit, mouse, frog, snake, hawk. Identify all food chains, the trophic levels of the hawk, and the effect of removing the frogs. 5

OR

Explain how a pesticide sprayed on crops can pass along a food chain: plants, insects, frogs, snakes. Suggest alternatives to chemical pesticides. 5

END OF QUESTION PAPER — Total: 40 Marks